

$$\begin{aligned} \text{Maximize } z &= 2x_1 + 3x_2 \\ \text{subject to } 2x_1 + x_2 &\leq 4 \\ x_1 + 2x_2 &\leq 5 \\ x_1, x_2 &\geq 0 \end{aligned}$$

$$\text{Maximize } z = 2x_1 + 3x_2$$

$$2x_1 + x_2 + s_1 = 4$$

$$x_1 + 2x_2 + s_2 = 5$$

$$x_1, x_2, s_1, s_2 \geq 0$$

$$m = 2$$

$$n = 4$$

$$4 - 2 = 2$$

↓

Iteration 1

$$\left\{ \begin{array}{l} x_1 = x_2 = 0 \text{ (N.B.)} \\ s_1 = 4, s_2 = 5 \text{ (B)} \end{array} \right\}$$

N.B.

$$x_1 = x_2 = 0$$

$$\Rightarrow s_1 = 4, s_2 = 5$$

fix to zero.

≥ 0 Basic.

$$z_0 = 0$$

Limiting Value(x_2)

$$4$$

$$- \dots \frac{5}{2} \dots \text{binding equation}$$

$$\min\left(\frac{5}{2}, 4\right) = \frac{5}{2}$$

$$\begin{array}{l} 0 \\ \uparrow \\ s_2 = 5 - 2x_2 \end{array}$$

$$\begin{aligned} s_1 &= 4 - x_2 \\ s_2 &= 5 - x_1 - 2x_2 \\ z &= 2x_1 + 3x_2 \end{aligned}$$

$z_0 \nearrow$ $x_1 \uparrow$ $x_2 \uparrow \Rightarrow x_2$ should be incr.

How much? $x_2 = 1000, x_2 = 100, x_2 = 10$??

(Entering Variable) $x_2 = \frac{5}{2}$ (becomes basic)

(Leaving Variable) $s_2 = 0$ (becomes N.B.)

Iteration 2

$$\left\{ \begin{array}{l} x_2 = \frac{5}{2}, s_1 = \frac{3}{2} \text{ (Basic)} \\ s_2 = 0, x_1 = 0 \text{ (N.B.)} \end{array} \right\} \rightarrow z_1 = 2 \cdot 0 + 3 \cdot \frac{5}{2} = \frac{15}{2}$$

$$2x_2 = 5 - x_1 - s_2$$

$$x_2 = \frac{5}{2} - \frac{1}{2}x_1 - \frac{1}{2}s_2$$

B.

N.B.

N.B.

N.B.

$$\overbrace{1 \quad 2 \quad 2 \quad 2}^{\text{B.}} \\ S_1 = 4 - 2x_1 - \underbrace{x_2}_{\text{N.B.}}$$

$$S_1 = 4 - 2x_1 - \left(\frac{5}{2} - \frac{1}{2}x_1 - \frac{1}{2}S_2 \right)$$

$$S_1 = \frac{3}{2} - \frac{3}{2}x_1 + \frac{1}{2}S_2$$

$$z = 2x_1 + 3x_2$$

$$z = 2x_1 + 3 \cdot \left(\frac{5}{2} - \frac{1}{2}x_1 - \frac{1}{2}S_2 \right)$$

$$z = \frac{15}{2} + \frac{1}{2}x_1 - \frac{3}{2}S_2$$

$$x_2 = \frac{5}{2} - \frac{1}{2}x_1 - \frac{1}{2}S_2$$

$$\star S_1 = \frac{3}{2} - \frac{3}{2}x_1 + \frac{1}{2}S_2$$

Limiting
value

5

$$\frac{1}{\min(1, 5)} = 1$$

$$x_1 \uparrow \quad z \uparrow \Rightarrow x_1: \text{E.V.}$$

$$x_1 = 1 \rightarrow \text{becomes Basic.}$$

L.V. $\Rightarrow S_1$ becomes N.B. in the next it.

Iteration 3

$$1 \quad 1 \quad x_1 = 2 \quad (\text{Basic}) \quad z_2 = 2 \cdot 1 + 3 \cdot 2$$

$$\left(\begin{array}{l} x_1 = 1, x_2 = 2 \quad (\text{Basic}) \\ S_1 = 0, S_2 = 0. \quad (\text{NB}) \end{array} \right) \quad z = 2 \cdot 1 + 3 \cdot 2 = 2 + 6 = 8.$$

① $x_1 = 1 + \frac{1}{3}S_2 - \frac{2}{3}S_1$

$$x_2 = \frac{5}{2} - \frac{1}{2}x_1 - \frac{1}{2}S_2$$

$$\frac{1}{6} - \frac{3}{6}$$

$$x_2 = \frac{5}{2} - \frac{1}{2} \left(1 + \frac{1}{3}S_2 - \frac{2}{3}S_1 \right) - \frac{1}{2}S_2$$

② $x_2 = 2 - \frac{2}{3}S_2 + \frac{1}{3}S_1$

$$z = \frac{15}{2} + \frac{1}{2}x_1 - \frac{3}{2}S_2$$

$$z = \frac{15}{2} + \frac{1}{2} \left(1 + \frac{1}{3}S_2 - \frac{2}{3}S_1 \right) - \frac{3}{2}S_2$$

③ $z = 8 - \frac{4}{3}S_2 - \frac{1}{3}S_1$

$$\frac{1}{6} - \frac{3}{6}$$

Is it possible to increase z ?

$S_1 \uparrow \quad z \downarrow \sim S_2 \uparrow \quad z \downarrow$ No
max.

So, the optimum solution is obtained.

$$S_1^* = 0, S_2^* = 0$$

$$x_1^* = 1, x_2^* = 2$$

$$z^* = 8$$

Algebraic form of Simplex.

Tabular form of Simplex

LP in eq. form.

Maximize $z = 2x_1 + 3x_2 \Rightarrow z - 2x_1 - 3x_2 = 0$

$2x_1 + x_2 + s_1 = 4$

$x_1 + 2x_2 + s_2 = 5$

$x_1, x_2, s_1, s_2 \geq 0$

$z - 2x_1 - 3x_2 = 0$

LHS = RHS

Variables (obj. function) constant

Basic variables = s_1, s_2

	Basic	z	x_1	x_2	s_1	s_2	R.H.S Solution	
	z	1	-2	-3	0	0	0	
s_1 -row	s_1	0	2	1	1	0	4	
s_2 -row (L.V.)	s_2	0	1	2	0	1	5	
	z	1	-1/2	0	0	3/2	15/2	
	s_1	0	3/2	0	1	-1/2	3/2	
New x_2 -row	x_2	0	1/2	1	0	1/2	5/2	

z $\uparrow$ x_2 : E.V.
min ratio $\rightarrow$ limiting values.
 $4/1 = 4$
 $5/2 = 5/2^*$

New z -row = (New x_2 -row) $\cdot$ (3) + old z -row.

3 15

2-10-

	0	$\frac{3}{2}$	3	0	$\frac{3}{2}$	$\frac{15}{2}$
P	1	-2	-3	0	0	0
	1	$-\frac{1}{2}$	0	0	$\frac{3}{2}$	$\frac{15}{2}$

New S_1 -row = (New X_2 -row) (-1) + S_1 -row

	0	$-\frac{1}{2}$	-1	0	$-\frac{1}{2}$	$-\frac{5}{2}$
	0	2	1	1	0	4
	0	$\frac{3}{2}$	0	1	$-\frac{1}{2}$	$\frac{3}{2}$

	2	1	$-\frac{1}{2}$	0	0	$\frac{3}{2}$	$\frac{15}{2}$	$X_1 = EV.$
LV. $\leftarrow$	S_1	0	$\frac{3}{2}$	0	1	$-\frac{1}{2}$	$\frac{3}{2}$	min ratio
row	X_2	0	$\frac{1}{2}$	1	0	$\frac{1}{2}$	$\frac{5}{2}$	$\frac{3/2}{1/2} = 1^*$
	2	1	0	0	$\frac{1}{3}$	$\frac{4}{3}$	8	$\frac{5/2}{1/2} = 5$
	X_1	0	1	0	$\frac{2}{3}$	$-\frac{1}{3}$	1	
	X_2	0	0	1	$-\frac{1}{3}$	$\frac{2}{3}$	2	

No EV.
6 pt. ✓

$$2 + \frac{1}{3}S_1 + \frac{4}{3}S_2 = 8$$

$$\begin{aligned} X_1^* &= 1 \\ X_2^* &= 2 \\ Z^* &= 8 \\ S_1 &= 0 \end{aligned}$$

$$x_2 + \frac{1}{3}s_1 + \frac{1}{3}s_2 = 0$$

$$s_1 = 0$$

$$s_2 = 0$$

$$\left(\begin{matrix} \text{New} \\ x_1\text{-row} \end{matrix} \right) \left(\frac{-1}{2} \right) + x_2\text{-row} = \text{New } x_2\text{-row}$$

$$\begin{array}{cccccc} 0 & 1/2 & 1 & 0 & 1/2 & s/2 \\ 0 & -1/2 & 0 & -1/3 & 1/6 & -1/2 \end{array}$$

$$\left(\begin{matrix} \text{New} \\ x_1\text{-row} \end{matrix} \right) \frac{1}{2} + \text{Old } z\text{-row} = \text{New } z\text{-row}$$

$$\begin{array}{cccccc} 0 & +1/2 & 0 & 1/3 & -1/6 & 1/2 \\ +1 & -1/2 & 0 & 0 & 3/2 & 15/2 \\ \hline 1 & 0 & 0 & 1/3 & 4/3 & 8 \end{array}$$

(-) coefficients in

z-row are the candi-

dates for E.V.

If z-row doesn't contain any (-) coefficient, then

MAX

any (-) coefficient /
you reached the opt.

Example 1:

$$\max z = 6x_1 + 5x_2$$

$$\text{s.t. } x_1 + x_2 \leq 5$$

$$3x_1 + 2x_2 \leq 12$$

$$x_1, x_2 \geq 0$$

$\Rightarrow$

$$\max z = 6x_1 + 5x_2$$

$$x_1 + x_2 + s_1 = 5$$

$$3x_1 + 2x_2 + s_2 = 12$$

$$x_1, x_2, s_1, s_2 \geq 0$$

Starting b.f.s = $(s_1, s_2) = (5, 12)$

Basic	x_1	x_2	s_1	s_2	RHS	
(max) z	-6	-5	0	0	0	min ratio
s_1	1	1	1	0	5	$5/1 = 5$
s_2	3	2	0	1	12	$12/3 = 4^*$
z	0	-1	0	2	24	min ratio
s_1	0	1/3	1	-1/3	1	$1/(1/3) = 3^*$
x_1	1	2/3	0	1/3	4	$4/(2/3) = 6$
(max) z	0	0	3	1	27	
x_2	0	1	3	-1	3	
x_1	1	0	-2	1	2	

No. E.V. So we found the optimum.

$$x_1^* = 2 \quad x_2^* = 3 \quad s_1^* = 0$$

$$s_2^* = 0$$

$$z^* = 27$$

$$s_2^* = 0$$

Example 2:

$$\min z = 2x_1 + 7x_2 - 2x_3$$

$$\text{s.t. } x_1 + x_2 + x_3 \leq 1$$

$$-4x_1 - 2x_2 + 3x_3 \leq 2$$

$$x_1, x_2, x_3 \geq 0$$

$$\Rightarrow \begin{aligned} x_1 + x_2 + x_3 + s_1 &= 1 \\ -4x_1 - 2x_2 + 3x_3 + s_2 &= 2 \end{aligned}$$

$$x_1, x_2, x_3, s_1, s_2 \geq 0.$$

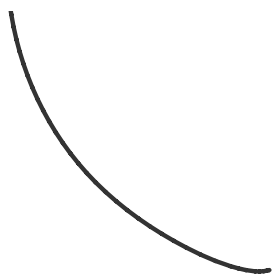
Starting b.f.s. $S = (s_1, s_2) = (1, 2)$

	Basic	x_1	x_2	(E.V.) x_3	s_1	s_2	RHS	
z		-2	-7	$2 \uparrow$	0	0	$2_0 = 0$	min Ratio
s_1		1	1	1	1	0	1	$1/1 = 1$
(L.V.) s_2		-4	-2	3	0	1	2	$2/3$
z		$2/3$	$-17/3$	0	0	$-2/3$	$-4/3 = -1.33$	
s_1		$7/3$	$5/3$	0	1	$-1/3$	$1/3$	$\Rightarrow$ L.V.
x_3		$-4/3$	$-2/3$	1	0	$1/3$	$2/3$	$1/3 \neq 0$
z		0	$-12/7$	0	$-2/7$	$-12/21$	$-10/7$	
x_1		1	$5/7$	0	$3/7$	$-1/7$	$1/7$	
x_3		0	$2/7$	1	$4/7$	$1/7$	$6/7$	

No positive value in z -row.

So we found the optimum

$$\left| \begin{aligned} x_1^* &= 1/7 & x_3^* &= 6/7 & z^* &= -10/7 \end{aligned} \right.$$



$$x_2^4 = 0$$

$$S_1^4 = 0$$

$$S_2^4 = 0.$$